Module 04: Cognition in Robotics

Learning Objectives

1. Define **Cognition** within the context of Robotics.
2. Analyze how Cognition interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Cognition.

Introduction

This module explores **Cognition**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Cognition is a critical component of the 8-part Active Inference spine, bridging the gap between Perception and Action.

Key Concepts

1. Cognition as a Markov Blanket Boundary

How does Cognition define the boundary between the agent and the environment?

2. Generative Models of Cognition

What parameters involved in Cognition must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Cognition drive the perception-action loop?

Applications

In Robotics, we see Cognition manifest in: * **Specific Example 1**: A Kalman smoother running on a batch of IMU and lidar data implements offline cognitive processing by jointly estimating the entire robot trajectory and map -- unlike the filter (which only estimates the current state), the smoother propagates information both forward and backward in time, resolving ambiguities that were unresolvable in the forward pass and producing a globally consistent belief state analogous to how deliberate reflection refines initial perceptual judgments. * **Specific Example 2**: An Unscented Kalman Filter (UKF) maintaining state estimates for a nonlinear robotic system (such as a fixed-wing UAV with aerodynamic drag) implements cognitive inference without linearization by propagating sigma points through the true nonlinear dynamics model, capturing second-order effects that the EKF's Jacobian linearization would miss; this richer cognitive processing reduces estimation divergence during aggressive maneuvers where linear approximations break down.

Conclusion

Understanding Cognition allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.